

## Solution Requirements

|                      |                                                                           |
|----------------------|---------------------------------------------------------------------------|
| <b>Date</b>          | 02 July 2026                                                              |
| <b>Team ID</b>       | XXXXXX                                                                    |
| <b>Project Name</b>  | HDI Predictor (Human Development Index Prediction Using Machine Learning) |
| <b>Maximum Marks</b> | 4 Marks                                                                   |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                                      | Priority (High/Medium/Low) |
|------|----------------------|------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication       | Users can securely access the HDI Prediction system using login credentials. | High                       |
| 2    | Authorization levels | Admin manages the dataset and model. Users can only predict HDI values.      | High                       |

|          |                                   |                                                                                                                    |        |
|----------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|--------|
| <b>3</b> | External interfaces               | Connects with CSV dataset, trained ML model (.pkl) , and Flask web application.                                    | High   |
|          |                                   |                                                                                                                    |        |
| <b>4</b> | Transactions processing           | Accepts user input, processes it through the ML model, and returns the predicted HDI value.                        | High   |
| <b>5</b> | Reporting                         | Displays predicted HDI result and allows viewing prediction history (if implemented).                              | Medium |
| <b>6</b> | Business rules, etc.              | All input values must be valid numeric values. Prediction is generated only after all required fields are entered. | High   |
| <b>7</b> | Compliance to laws or regulations | Protect user data and follow general data privacy and security guidelines.                                         | Medium |

|   |       |                                                                                |        |
|---|-------|--------------------------------------------------------------------------------|--------|
| 8 | Other | System provides accurate and fast HDI prediction with a simple user interface. | Medium |
|---|-------|--------------------------------------------------------------------------------|--------|

**Step 2: Non-Functional Requirements (NFR)**

|   |                            |                                                               |                                   |
|---|----------------------------|---------------------------------------------------------------|-----------------------------------|
| 1 | Performance & Speed        | Prediction should be generated quickly.                       | Response time less than 2 seconds |
| 2 | Scalability                | System should support increasing users and larger datasets.   | Supports future expansion         |
| 3 | Security & Data Privacy    | User data and prediction requests should be handled securely. | Secure login and protected data   |
| 4 | Reliability & Availability | System should be available with minimal downtime.             | 99% availability                  |
| 5 | Usability & Accessibility  | Interface should be simple, responsive, and easy to use.      | User-friendly design              |

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|------|--------------|-------------------------|-------------------------------------|
|------|--------------|-------------------------|-------------------------------------|

|   |              |                                                          |                                 |
|---|--------------|----------------------------------------------------------|---------------------------------|
| 6 | <i>Other</i> | Maintainable, portable, and easy to update the ML model. | Easy deployment and maintenance |
|---|--------------|----------------------------------------------------------|---------------------------------|